

SOVOL

SOVOL ZERO

User Manual

V1.0



Dear customers:

Thank you for choosing Sovol printers!

Sovol is committed to providing excellent machines to 3D printing enthusiasts all over the world .This manual is designed for SOVOL ZERO owners to start their SOVOL ZERO printing journey.We still recommend all the SOVOL ZERO owners to read the manual carefully even if youare familiar with the 3D Printing technology, as there are lots of important information about the SOVOL ZERO for you to learn and help you get better printing exprience .In this manual there are some tutorials can be found on official website and group, you can scan the QR-codes.



sovol support group



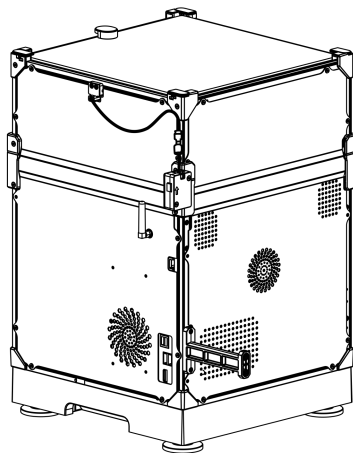
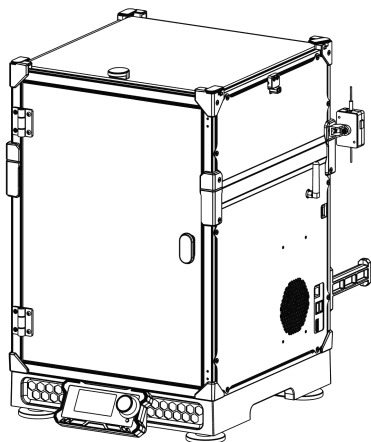
sovol official website

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Note

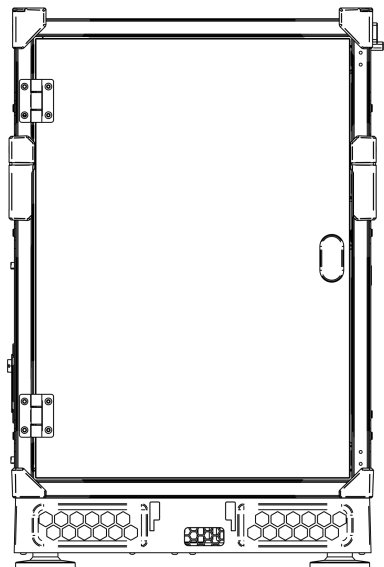


- Do not use the printer any way other than described here in order to avoid personal injury or property damage.
- Do not place the printer in the environment with large vibration or other instability. The shaking of the machine will affect the printing quality of the printer.
- Please do not place the machine in inflammable and explosive materials or near high heat source.
- Please place the machine in a ventilated, cool and dust free environment.
- It is recommended to use the material recommended by the manufacturer to avoid machine damage.
- Do not use any other power cable except the one supplied. Always use a grounded three-prong power outlet.
- Please do not open the plastic cover during usage, otherwise the printing will be interrupted.
- Do not wear cotton gloves when operating the printer. Such cloths may become tangled in the printers moving parts leading to burns, possible bodily injury, or printer damage.
- Please wait a moment to remove the print after the print is finished.
- It's not recommended to use the third party firmware or mainboard etc, or the warranty will be void.
- Clean the printer frequently. Always turn the power off when cleaning, and wipe with a dry cloth to remove dust, adhered printing plastics or any other material off the frame, guide rails, or wheels. Use glass cleaner or isopropyl alcohol to clean the print surface.
- Children under 10 years should not use the printer without supervision.
- This machine is equipped with a security protection mechanism. Do not manually move the nozzle and printing platform mechanism manually while booting up, otherwise the device will automatically power off for safety.
- Users should comply with the laws and regulations of the corresponding countries and regions where the equipment is located (used), abide by professional ethics, pay attention to safety obligations, and strictly prohibit the use of our products or equipment for any illegal purposes. Sovol will not be responsible for any violators' legal liability under any circumstance.
- Regularly clean the rails and leadscrews, and apply grease for maintenance.

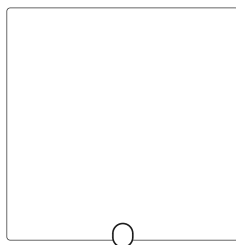


Model	SOVOL ZERO
Software language	English
Print method	Network interface USB cord and WIFI
Type	FDM
Number of nozzles	1
Print size	152.4*152.4*152.4mm
Recommend Printing speed	≤500mm/s
Printing accuracy	±0.1mm
Nozzle diameter	0.4mm (Replaceable)
Nozzle temperature	≤350°C
Hot bed temperature	≤120°C
Applicable filament	PLA/ABS/PETG/TPU
Diameter of filament	1.75mm
File format supported	G-code
Voltage	AC 100~120V / 200~240V, 50/60Hz,
Operation system	Windows, Mac
Power supply	150W/24V

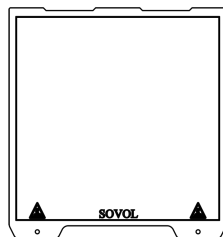
Package List



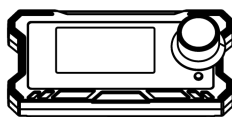
Printer



Top Glass Cover



PEI Plate



Knob Screen

Tool Box



Gasket



Isolation column



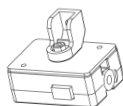
PTFE Tubing



Spool Holder



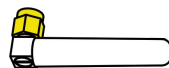
M3X16*3



Filament Detector



Filter protection cover



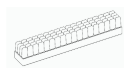
WiFi Antenna



Power cord



USB Flash Disk



Silicone brush



Scraper knife



Toolkit



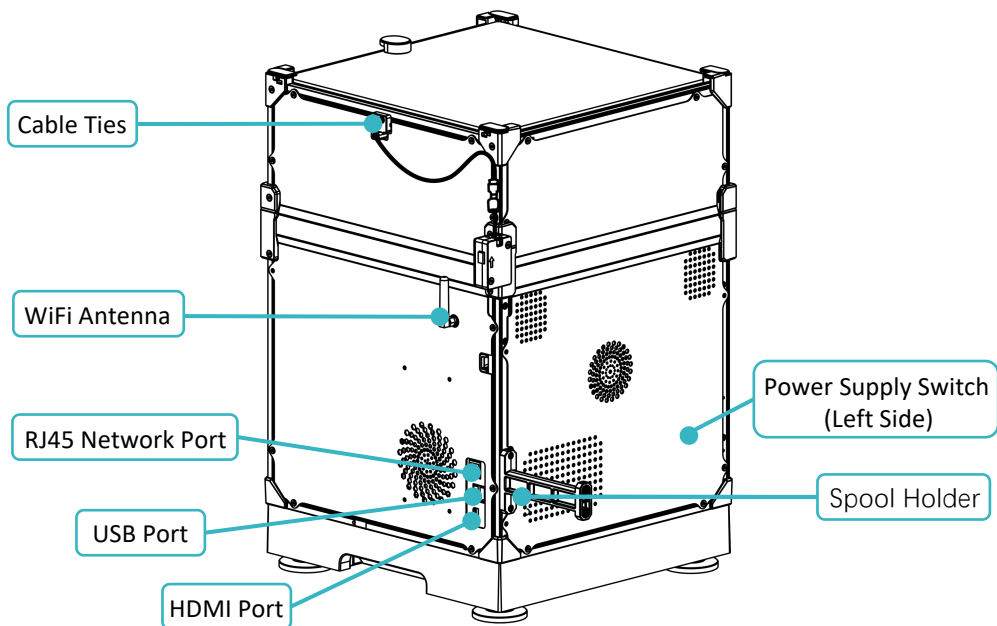
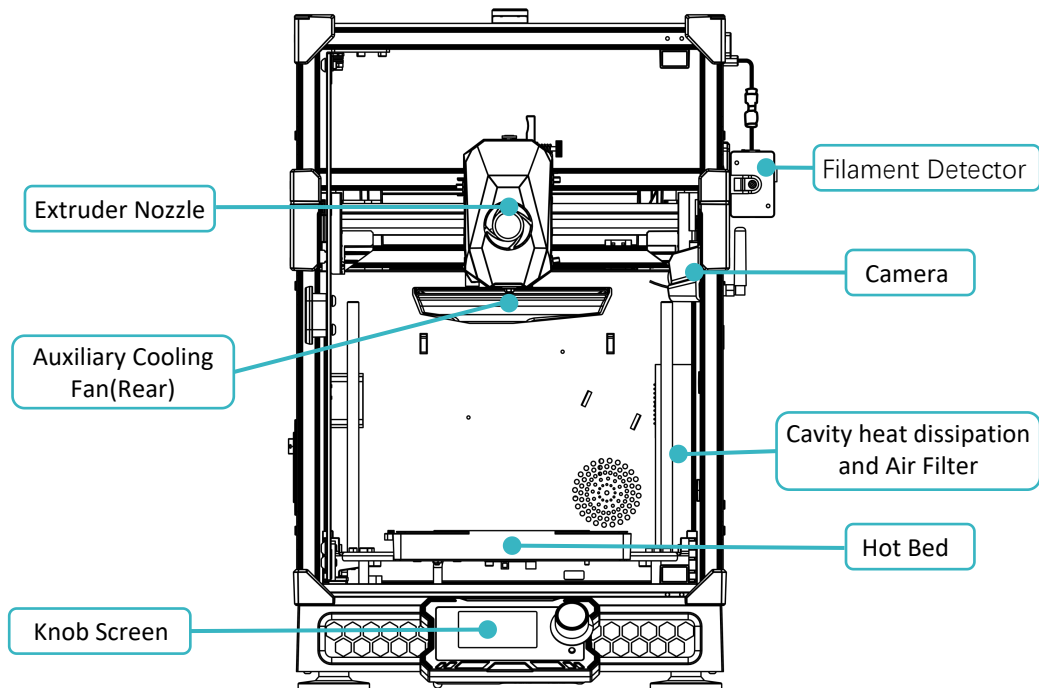
Diagonal pliers



Brass and Hardened Steel Nozzle



Nozzle Cleaner and Zip Tie



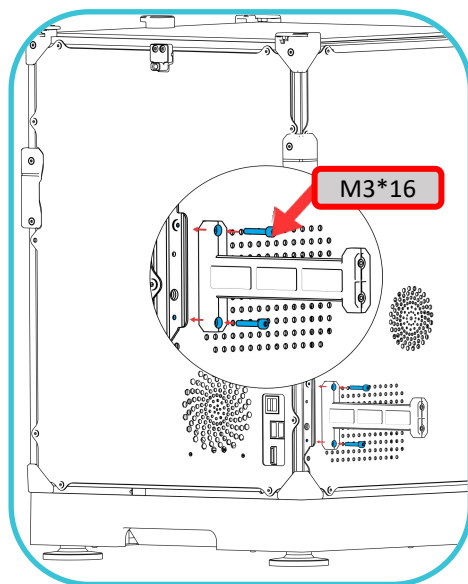
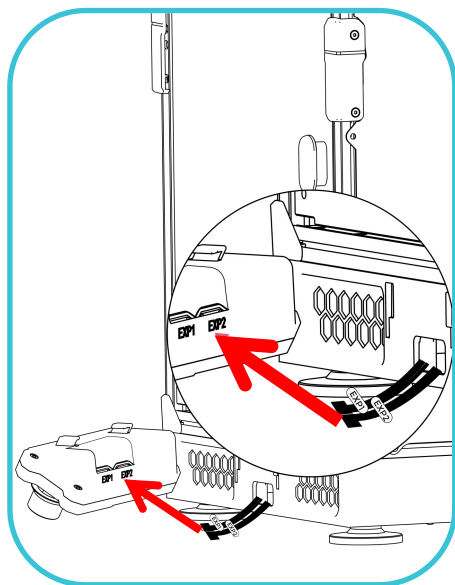
Unboxing

Unpack the outer package, take the printer out of the box, and place it on the desk. After removing all the foam, put the other accessories aside, and follow the steps below for installation.

STEP 01

Screen Installation:

There are two screen cables directly below the machine. Insert each cable into its corresponding hole in sequence, then snap the screen into the designated slots at the bottom.



STEP 02

Installation of the rack:

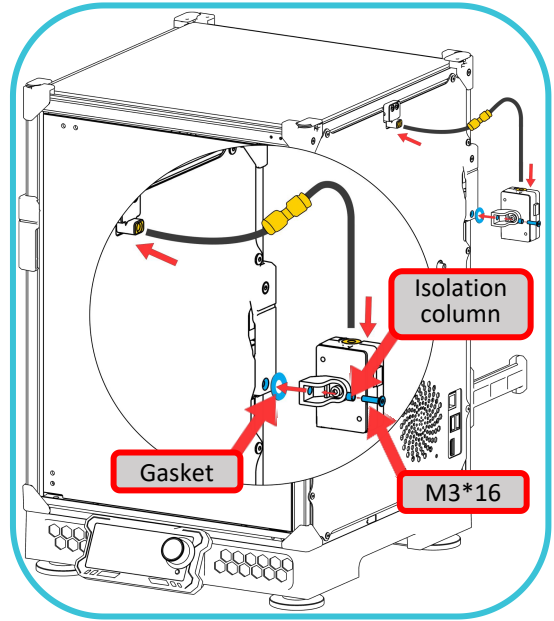
First, insert two M3X16 screws into the corresponding holes at the lower left corner and tighten them.

Unboxing

STEP 03

material breakage detection :

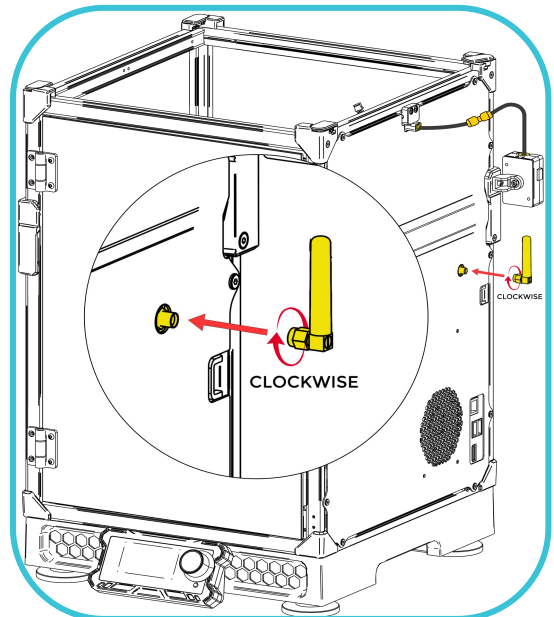
First, the PTFE tube on the extrusion nozzle should be threaded out of the fixed bracket (as shown), inserted into the material breaking detection, and then the M3x16 screw in the kit should be put into the pre-lock hole and locked.



STEP 04

Installation of WiFi antennas :

Take out the upper half of the WiFi antenna, insert the corresponding hole and tighten it in a clock-wise direction.

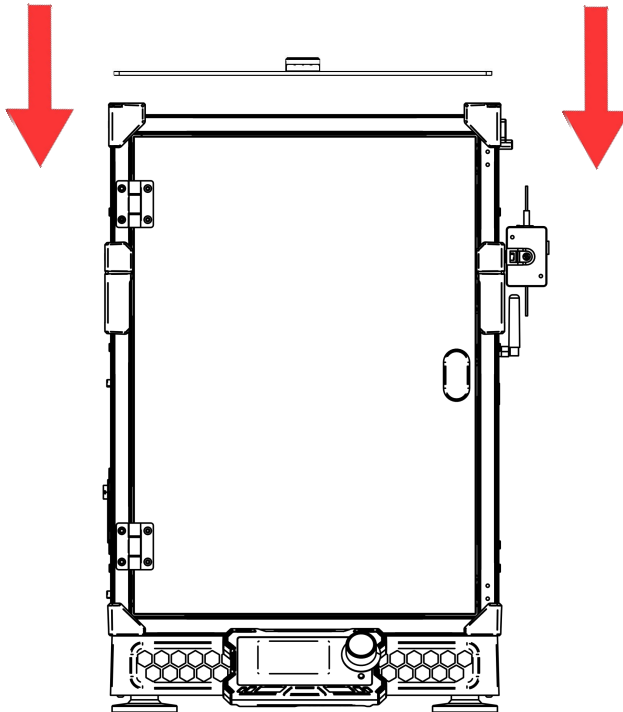
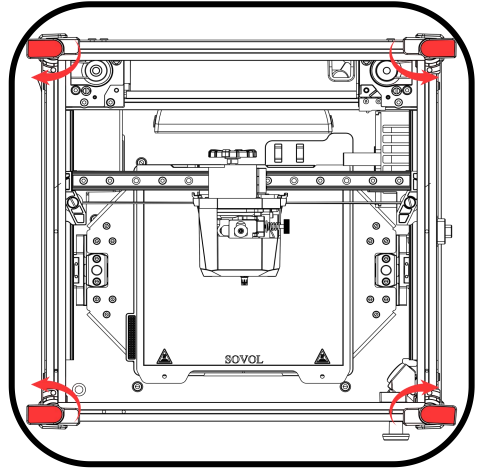


STEP 05

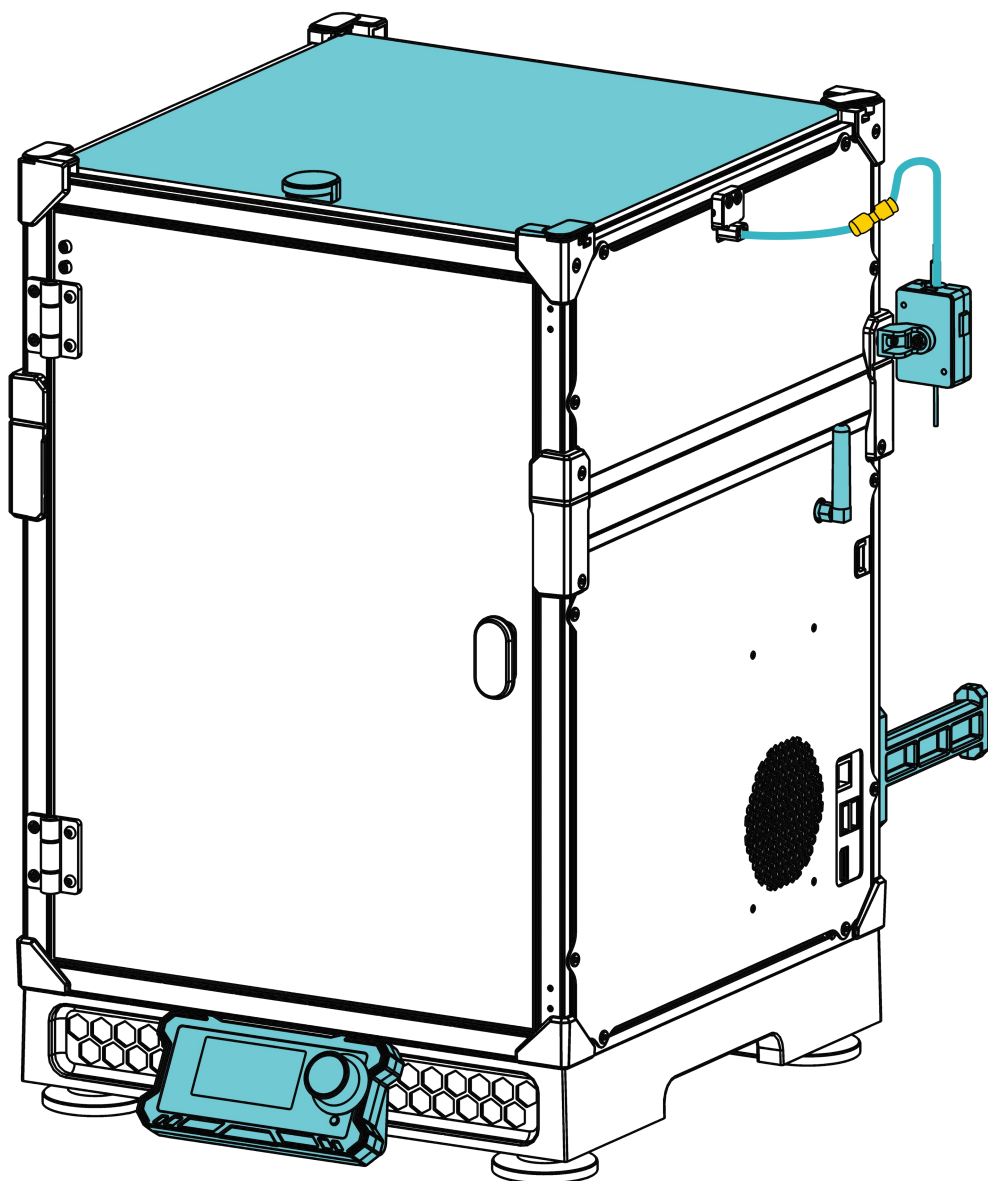
Installation of glass cover:

We turn the top corner of the machine, turn it outward (as shown), cover the glass cover and turn it back to complete the installation.

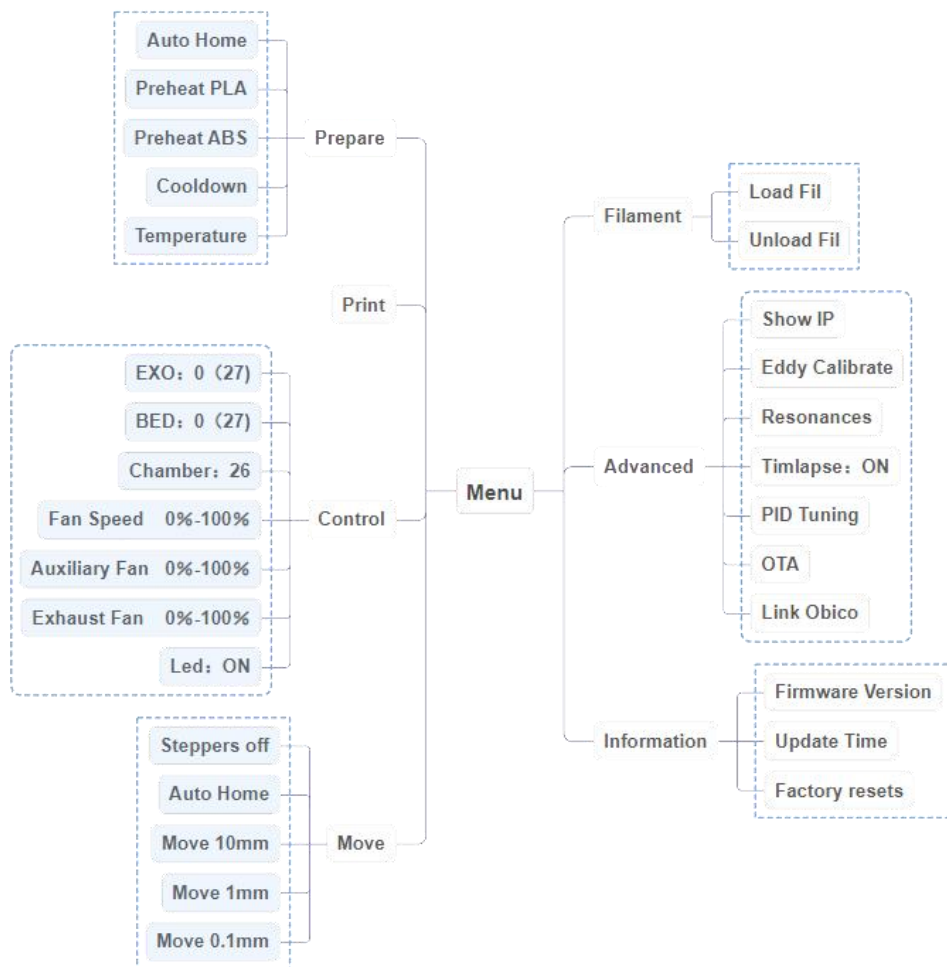
Finally, The PEI plate needs to be placed on the hot bed first.



PEI plates need to be cleaned regularly to ensure that the surface is grease-free.



Final renderings



Updates may differ, please refer to the latest firmware version. If you have any questions during use, please contact us, and we will respond as soon as possible.

Mian



Note: The current interface is for reference only, due to the continuous upgrading of functions, the actual latest software/firmware UI on the official website shall prevail.

1. Nozzle temperature: Displays the current temperature/preset value temperature of the nozzle

2. Heating bed temperature: Displays the current temperature/preset value temperature of the heating bed

3. Printing progress bar: displays the printing progress percentage, starting printing 0% - printing completed 100%

4. Printer status: There will be four states appearing at this time.

- ① Ready : The machine is ready**
- ② Power Recovery : Power-resume print function**
- ③ Factory Settings : Factory reset**
- ④ TiP code : Knob screen code**

5. Fan rotation speed percentage: displays the cooling fan rotation speed percentage, 0%-100%

6. Printing speed: displays the current printing speed. Printing speed can be adjusted with the knob.

7. Print progress time: Shows how long the print model has been completed.

Start-up

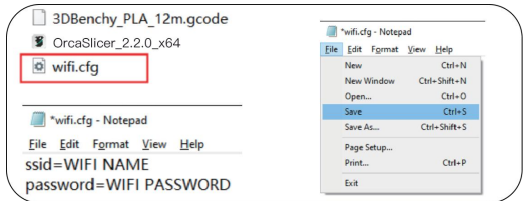
1. Use the USB flash drive included in the machine's accessory package and insert it into your computer to access its contents. Search for the file named "wifi.cfg" within the drive.

2. Open the "wifi.cfg" file using Notepad; this action resembles the example displayed on the right. Enter your WiFi network name (SSID) and password in the following format:

- ssid=your_WiFi_network_name
- password=your_WiFi_password

Example:

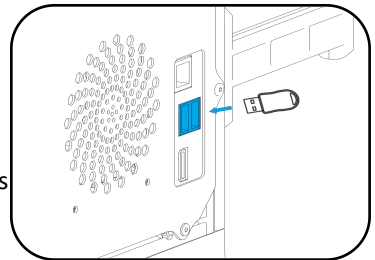
- ssid= WiFi NAME
- password= WiFi Password



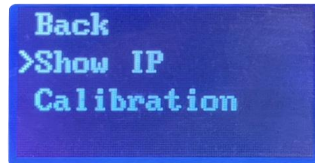
3. After entering your WiFi info, save the "wifi.cfg" file, ensuring it remains in the USB's root directory.

*Please make sure you type in the correct WiFi info.

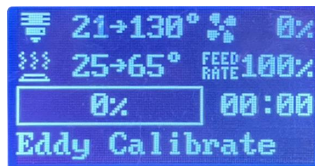
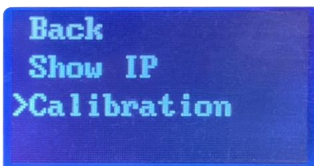
4. Power on the machine, enter the system and then plug the USB port into one of the two USB ports on the right side of the machine. Once inserted, pause for 15-20 seconds before continuing.



5. Press the knob to select 'Show IP' and wait for 5-10s. The IP address of your home network will appear. If 127.0.0.1 appears, we need to check if the Wi-Fi account and password in the USB drive's files are correct, or if the USB drive is properly inserted, and then restart the machine to perform the above actions again.



6. After the machine is connected to Wi-Fi, press the knob again to select the auto-calibration function, the machine will automatically calibrate and then restart.



Enable camera

Attentions:

1. WIFI must use 2.4G band signal.
2. Pay attention to the strength of the WIFI signal, the router connected without obstacles in a. The straight line distance from the machine should not exceed 10m.
3. Make sure the wifi name and password are correct.
4. computer or cell phone must be used with the printer is the same network wifi
5. If the printer display does not show the IP address, or show the IP address "172.0.0.1", please recheck the wifi. Please recheck the name and password in the wifi.cfg file to make sure they are correct. And try "show IP" several times.
6. If you want to change the WiFi network, please re-enter the name and password of the WiFi network you want to connect to in the wifi.cfg file. If you want to change the WiFi network, please re-enter the name and password in the wifi.cfg file.

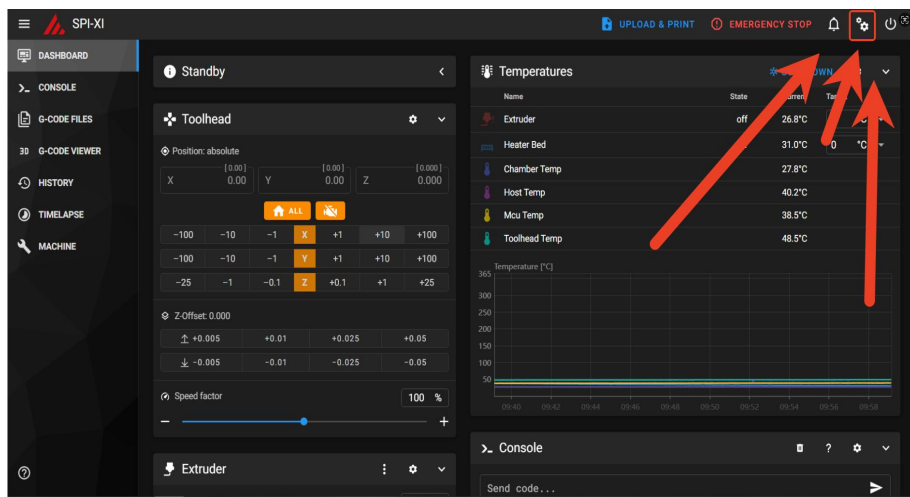


In the event of a missing configuration file, follow these quick steps to generate a new one:

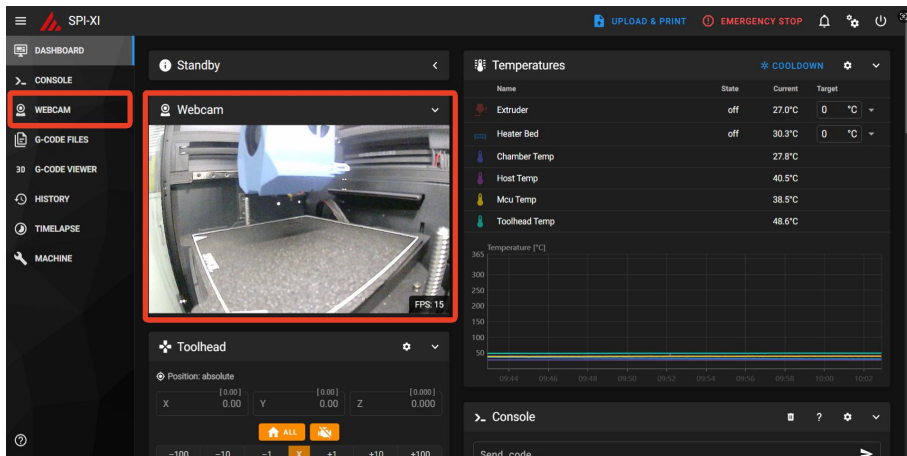
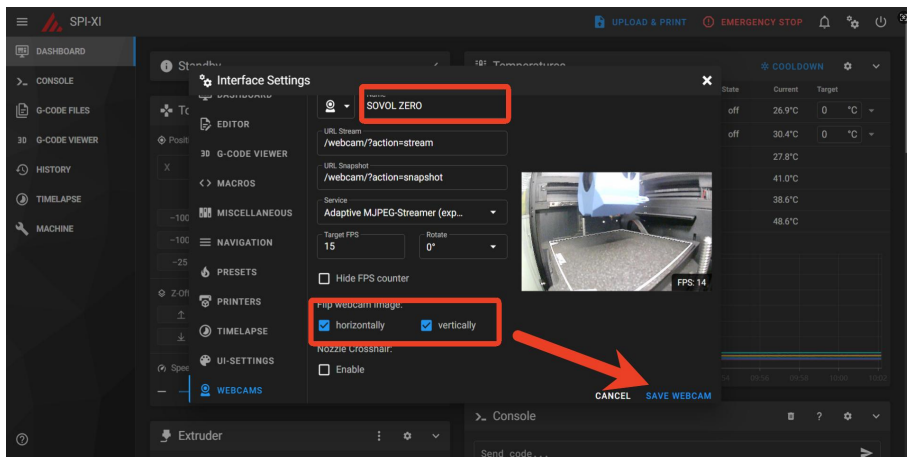
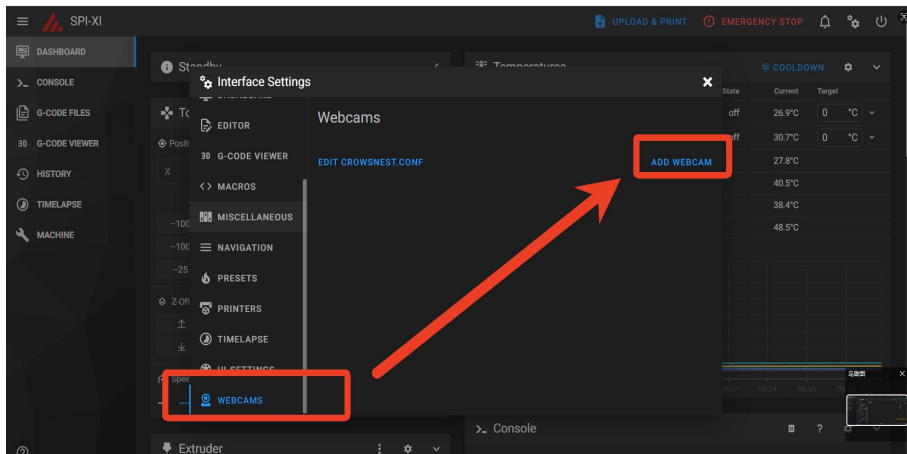
1. Right-click on your desktop and select "New" > "Text Document" to create a new TXT file.
2. Rename this file to "wifi.cfg" with a lowercase "cfg" extension.
3. Confirm the file extension change by clicking "Yes" when prompted, effectively creating your new .cfg file.

Enable camera

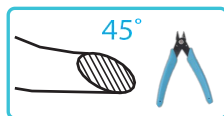
After the machine successfully connects to the network, operate "Show ip" on the screen to obtain the machine's network IP. We need to enter the IP address in the web interface within the same LAN to log in to mainsail (accessible from both mobile devices and computers). Please follow the steps below to enable the camera.



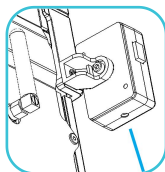
Enable camera



Fill Filament

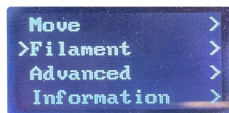


Use the scissors in the accessory kit to cut off the end of the filament at a 45° angle.



Steps to insert filament: Insert the filament into the hole of the filament runout sensor until it reaches the end. When the consumables cannot be inserted any further, complete the step of inserting filament.

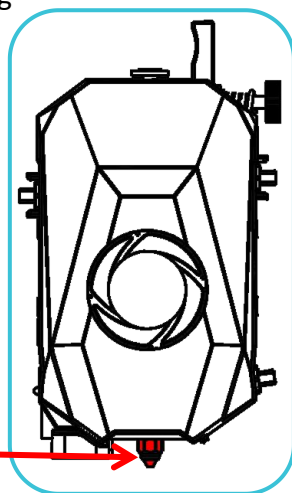
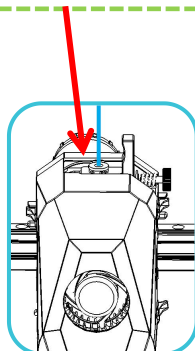
Make sure the filament are loaded correctly & are engaged with the extruder gears.



Select the "Load Fil" function, the heating block will heat up, reach the specified temperature value, the E-axis extrusion motor starts to rotate, the E-axis motor rotation process can check whether the consumables have been loaded, confirm the extrusion normal, the loading of consumables step is completed.

After completing the consumables procedure, if no consumables appear in the nozzle, repeat the preceding operations.

After the nozzle is loaded with the required filament, it is essential to check if there is any leftover filament on the nozzle. If there is leftover filament on the nozzle, the return to zero reset will be inaccurate, and the nozzle will scratch the PEI plate, which damages both the nozzle and the PEI plate, significantly reducing print quality and performance.



Please keep it clean

Slicing Software



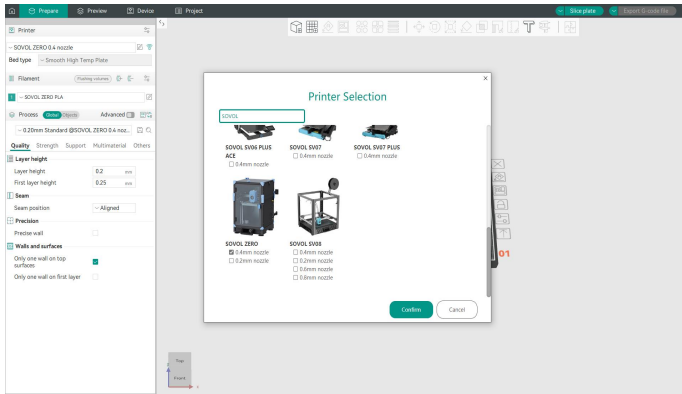
The latest version in official website shall prevail



OrcaSlicer_2.2.0_x64

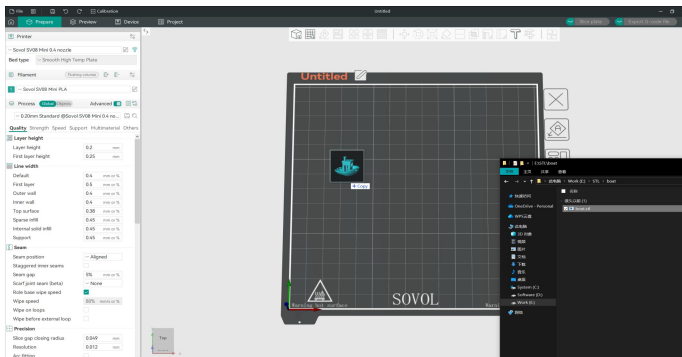
1

Files installed in the USB “OrcaSlicer”.



2

Select add “SOVOL ZERO”



3

Select and add the model you want to print.

Back
Prepare
>Print
Control

Back
>USB-boat.stl.gc

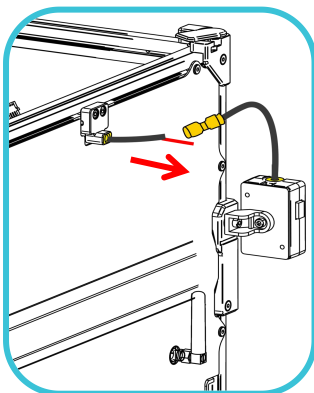
You can view the Show
Ip during printing

Factory Settings

- During the 3D printing process, if the filament runs out, the machine will trigger the material break detection warning, and we need to unload the residual filament.



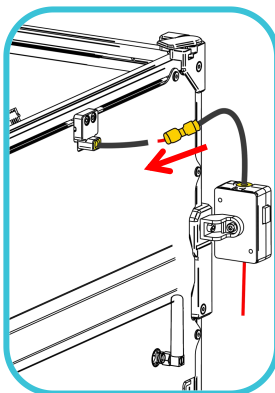
2



During the unloading process, the air tube connector on the front end needs to be removed, and the remaining filaments should be taken out.

When loading new filaments, you need to feed the filaments through the material breaking detection to the extrusion wheel of the nozzle, then select to load the filament. Once filament is extruded from the nozzle, select to continue printing; if the filament fails to extrude, release the extruder handle, remove the filament, and repeat the above steps.

3



```

Back
Tune      >
>Change fil  >
Resume printing
    
```

```

Back
>Load Fil
Unload Fil
    
```

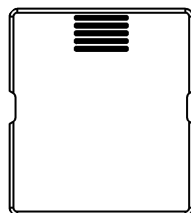
```

Back
Tune      >
>Change fil  >
>Resume printing
    
```

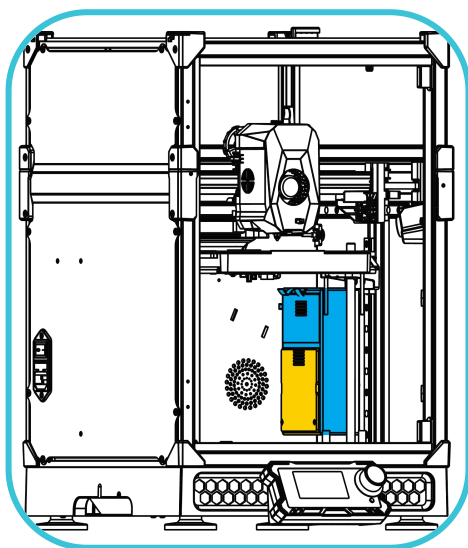
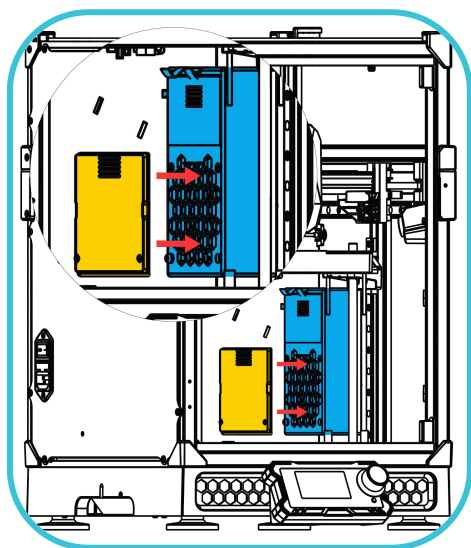
Air filtration system installation

When you need to print high-temperature filaments such as ABS, PC, CF, nylon, etc., you need to take out the filter protection cover in the accessory pack, align the hole on the air filter and install it

【As shown in the following picture】 to start printing.



Filter protection cover

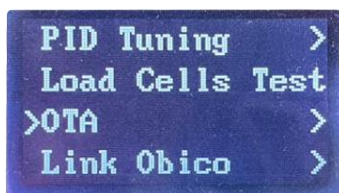


OTA upgrade:

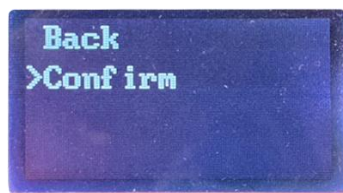
When our firmware version is at a lower version, please update the firmware version in a timely manner. This update can be completed by operating on the device itself.



①.Click on 'Advanced'



②.Click on 'OTA'



③.Click



- 1 The system will check for updates, wait for the updates to complete, and the machine will automatically restart. After the update, please power off and restart.



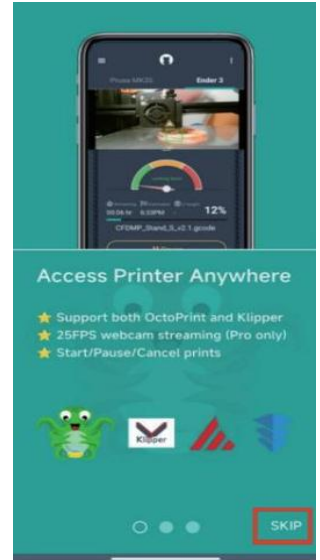
- 2 During system checks, if it is already the latest version, "latest version" will be displayed.

Obico Download

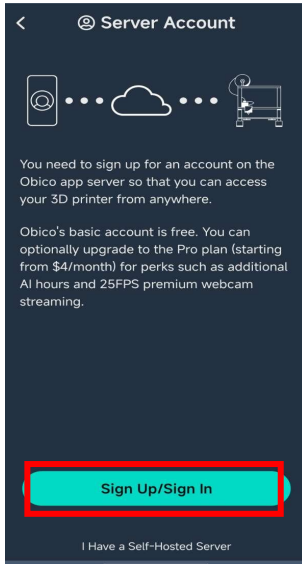
- For Android users: visit the Google Play store, search for "Obico" and select install app.
- For iOS users: Visit the Apple APP Store, search for "Obico", and then select Install APP.
- Via Web interface: If you want to access via Web interface

<https://obico.io>

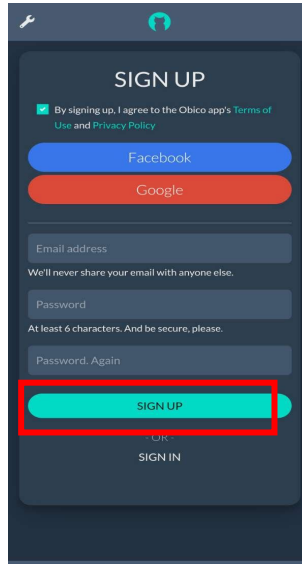
- Start the Obico operating interface of the printer and follow the steps below.
 - Connect the Obico software on your phone.
- Open Obico downloaded from the above website



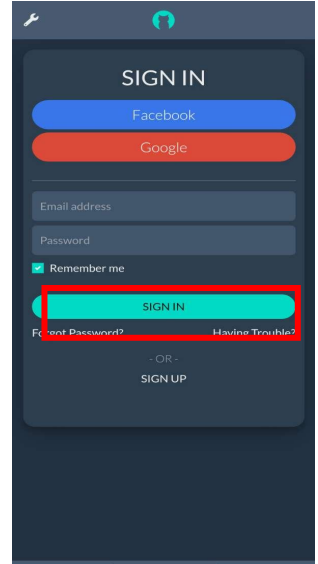
①. Click on 'SKIP'



②. Click on 'Sign Up/Sign In'



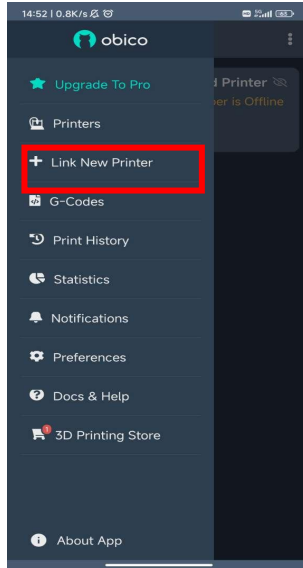
③. Register an account, enter the information, and click on 'SIGN UP'



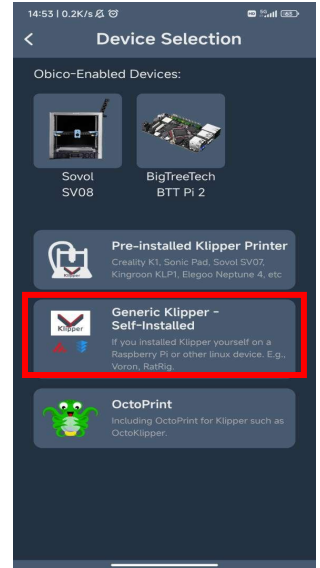
④. Enter the account information and click on 'SIGN IN'



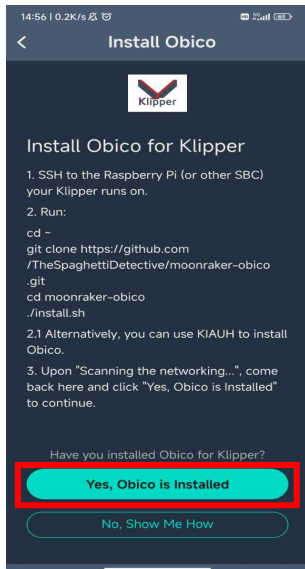
⑤. Click on the menu page



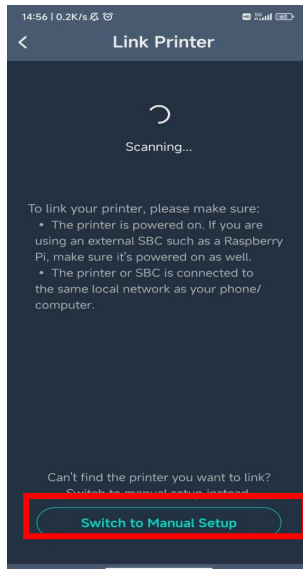
⑥. Link New Printer



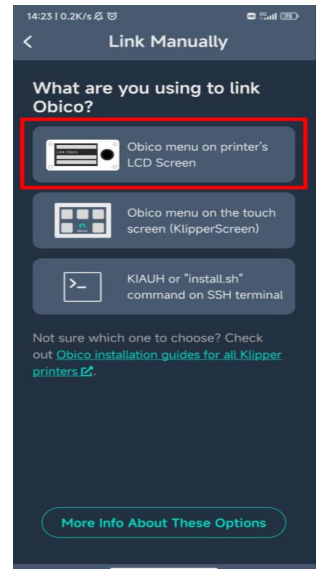
⑦. Click



⑧. Click

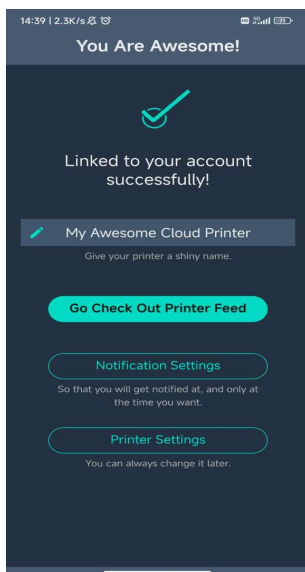
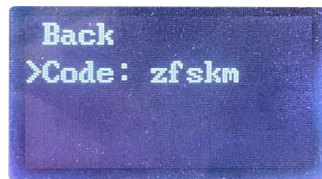
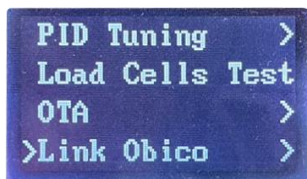
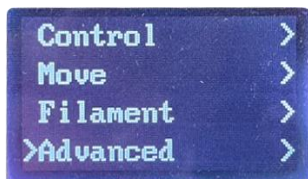


⑨. Click



⑩. Click

Select the obico function on the machine and a 5-digit verification code will appear.

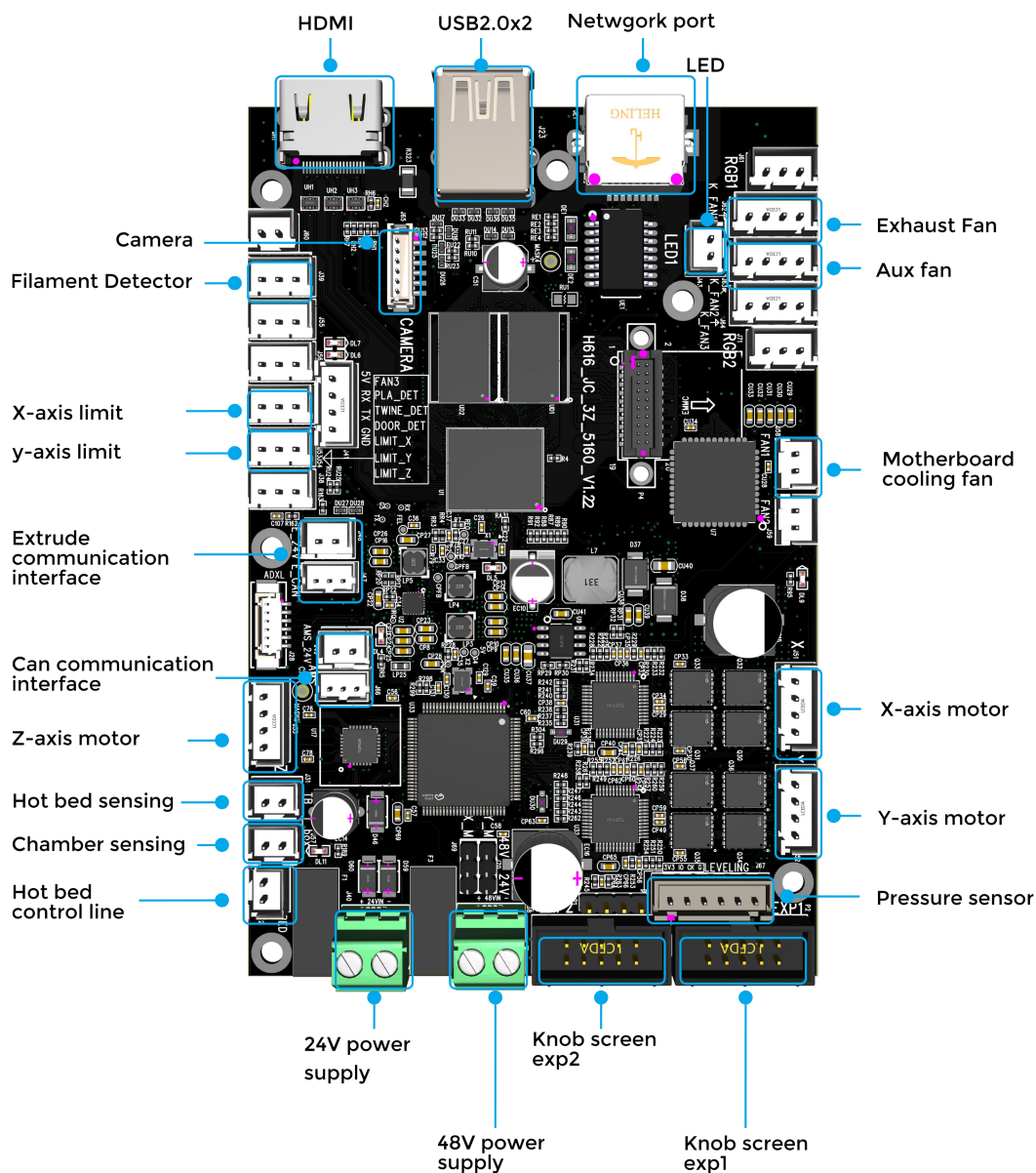


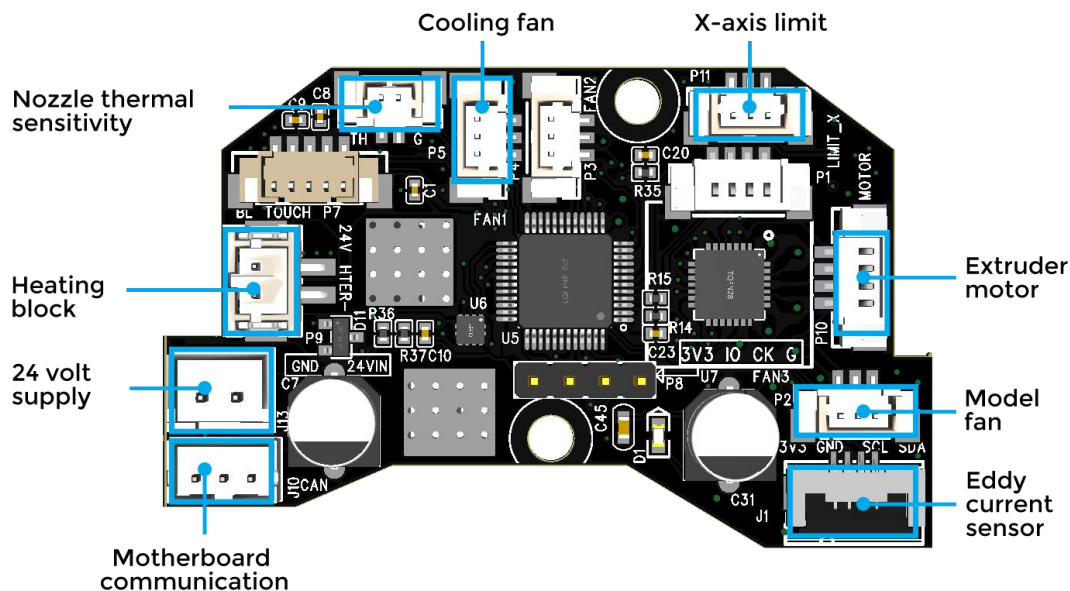
⑪ Printer connected successfully



Updates may differ, please refer to the latest firmware version. If you have any questions during use, please contact us, and we will respond as soon as possible.

Motherboard







sovol support group



sovol official website

SHENZHEN Liandianchuang Technology CO., LTD.

Official Website: sovol3d.com E-mail: info@sovol3d.com

